

HOUSE PRICE PREDICTION USING LINEAR REGRESSION

Artificial Intelligence & Machine Learning – Internship Task 1

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1. Introduction

The objective of this project is to build a machine learning model capable of predicting house prices using the California Housing Dataset. Linear Regression was chosen as the baseline algorithm because of its simplicity, interpretability, and effectiveness for regression tasks. This project demonstrates the complete machine learning workflow from data loading to evaluation.

2. Dataset Description

The California Housing Dataset is available through Scikit-Learn. The dataset contains housing-related attributes such as Median Income, House Age, Average Rooms, Population, Latitude, and Longitude. The target variable represents the median house value. The dataset is widely used for learning regression techniques and model evaluation.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand the structure and quality of the dataset. The following activities were completed: • Dataset shape inspection • Feature analysis • Missing value detection • Statistical summary generation • Correlation analysis • Data visualization through heatmaps and histograms Key Observation: Median Income typically shows the strongest positive relationship with house prices.

4. Model Development

The dataset was divided into training and testing sets using an 80:20 split. A Linear Regression model was trained on the training data. The model learned relationships between input features and the target variable. Predictions were then generated on the test dataset.

5. Model Evaluation

The model was evaluated using: • Mean Absolute Error (MAE) • Root Mean Squared Error (RMSE) • R^2 Score These metrics help determine the prediction accuracy and overall effectiveness of the model. Results (Replace with notebook output): MAE = _____ RMSE = _____ R^2 = _____

6. Results and Discussion

The Linear Regression model provides a solid baseline for predicting housing prices. A higher R^2 Score indicates better explanatory power, while lower MAE and RMSE values indicate improved prediction accuracy. Graphs such as Actual vs Predicted values can be used to visually assess model performance.

7. Future Improvements

Possible enhancements include: • Feature Engineering • Hyperparameter Optimization • Cross Validation • Random Forest Regressor • Gradient Boosting • XGBoost • Ensemble Learning Techniques

8. Conclusion

This project successfully demonstrated the complete machine learning lifecycle. The California Housing Dataset was analyzed, visualized, and used to train a Linear Regression model. The project provided practical experience with Python, Pandas, Data Visualization, Scikit-Learn, and evaluation metrics used in real-world machine learning applications.

9. References

1. Scikit-Learn Documentation 2. California Housing Dataset Documentation 3. Python Data Science Libraries (Pandas, NumPy, Matplotlib, Seaborn) 4. Machine Learning Learning Resources provided in Internship Task 1